

Beyond the Kilometer: Network-Revealed Healthcare Access in Brazil

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Abstract

Geographic distance is the standard metric for measuring healthcare access, yet it ignores how patients actually traverse the hospital network: through referrals, informal flows, and transport modes that roads do not encode. We build a bipartite graph of 5,588 Brazilian municipalities and 6,710 public hospitals from 179.5 million federal SUS hospital admissions (2010–2024), and learn 128-dimensional node embeddings via a sparse-on-the-fly node2vec implementation. Defining each municipality’s network-revealed isolation as the embedding distance to its nearest high-complexity hospital—and comparing it to the haversine distance to the same set—we find systematic and geographically coherent divergence in roughly half of all municipalities. The Northern region of Brazil is more connected than the map suggests, as river and air transport funnel patients to the Manaus and Boa Vista hubs along corridors invisible to road-based metrics. Conversely, the Northeastern semi-arid hinterland is more isolated than the map suggests, as patients bypass underequipped local hospitals and travel to distant regional capitals. The headline divergence map and a structured taxonomy of canonical cases provide a diagnostic of where the Brazilian SUS network deviates from its planning geography. We discuss why the cross-sectional relationship between this divergence and amenable mortality is confounded by SIM under-reporting and cross-border death recording, and outline a closure event-study design as the natural next step toward causal identification of the marginal mortality cost of network distance.

Keywords: hospital access, healthcare deserts, graph embeddings, node2vec, amenable mortality, SUS, Brazil.

JEL: I11, I14, I18, R12, C45.

*Insper. Email: darcio.g.martins@gmail.com. Submission to ETH/UZH Workshop in AI & Applied Economics (May 2026). Replication code and data documentation: <https://github.com/darciogm/bitter-pills/tree/main/paper18-hospital-deserts>.

1 Introduction

Distance to the nearest hospital is one of the most widely used measures of healthcare access in applied economics. It motivates the literature on hospital closures (Avdic, 2016; Gujral and Basu, 2019; Carroll, 2019), on rural healthcare deserts (Germack et al., 2019; Lindrooth et al., 2018), and on the spatial allocation of health resources (Gaynor and Town, 2011). Empirical work uses haversine kilometers, road travel time (Doyle, 2011), or driving time over OpenStreetMap as inputs to demand systems, instrumental variables for utilization, or exposure measures in event studies of facility entry and exit. Yet geographic distance is, at best, a noisy proxy for the medically meaningful concept of *network access*—the actual ease with which patients reach the appropriate level of care.

This gap between geography and access is particularly stark in Brazil, where the public Sistema Único de Saúde (SUS) covers $\approx 75\%$ of the population through a hub-and-spoke regionalization built atop a 5,570-municipality administrative grid. Many patients in Amazonian municipalities reach high-complexity care exclusively by river boat or chartered plane; many in the semi-arid Northeast bypass nominally nearest hospitals because those facilities lack qualified specialists, ICU beds, or suitable equipment. In both cases, the metric “kilometers to the nearest hospital” fundamentally misses the structure of realized access. The error is not random: it is concentrated in the poorest and most epidemiologically vulnerable regions of the country.

This paper. We provide a graph-based, data-driven measure of healthcare access for the universe of Brazilian municipalities, and use it to map where the geographic and the network-revealed views of access agree and where they sharply diverge. Our object of analysis is the bipartite patient flow graph linking 5,588 municipalities of residence to 5,828 public hospitals through ≈ 100 million federal hospital admissions (AIH) over 2010–2024, the entire universe of SUS inpatient activity in the period. We learn a 128-dimensional representation of each node using a sparse implementation of node2vec (Grover and Leskovec, 2016) that encodes second-order random walks with edge-weight bias. Two embeddings are produced: a bipartite embedding in which municipalities and hospitals share a common space (allowing direct municipality-to-hospital cosine distance), and a TF–IDF-weighted co-patency projection of the bipartite graph onto the municipality-only space.

For each municipality, we compute two scalar measures of isolation relative to the set of high-complexity hospitals—those with active SUS habilitation in cardiology, oncology, neurology, nephrology, perinatology, trauma-orthopedics, or transplantation:

$$\text{iso}_i^{km} = \text{haversine km to nearest high-complexity hospital}, \quad \text{iso}_i^{emb} = 1 - \cos(\mathbf{e}_i, \mathbf{e}_{h^*(i)}),$$

Standardizing each to a z -score and taking the difference yields the *divergence* $\tilde{\Delta}_i = z(\text{iso}_i^{emb}) - z(\text{iso}_i^{km})$. A positive divergence indicates that the embedding sees a municipality as *more* isolated than the map does—patients leapfrog the nominally nearest hospital because it is inadequate. A

negative divergence indicates the opposite: the geography overstates isolation because patients reach a distant hub via channels (river, air, informal referral pathways across state lines) that road geography ignores.

Findings. Three patterns emerge.

First, the divergence is large. The cross-sectional standard deviation of $\tilde{\Delta}$ is 0.91, with 9% of municipalities exhibiting $|\tilde{\Delta}| > 1.5$. Roughly 56.7% of municipalities have a different nearest high-complexity hospital under the embedding metric than under the geographic metric.

Second, the divergence is geographically organized into recognizable macro-regional patterns. Northern (Amazonian) Brazil is dominantly *negatively* divergent: $\tilde{\Delta}_N = -0.22$, with median geographic distance to the nearest high-complexity hospital of 30 km but embedding-revealed connections that pull these municipalities toward Manaus, Boa Vista, and Belém despite distances often exceeding 200 km by air or river. Northeastern semi-arid municipalities show the opposite signature in pockets of Piauí, Maranhão, and Bahia interior: nominal hospital proximity (medians ≈ 19 km) coexists with high embedding-isolation, as patients route around inadequate local facilities to reach state-capital tertiary care. The Southeast and Center-West show positively-skewed but moderate divergence, consistent with metropolitan saturation effects in Greater São Paulo and the Brasília corridor. The South emerges as the most well-calibrated region: geographic and network-revealed access are closely aligned, with $\tilde{\Delta}_S = -0.10$ and the lowest cross-sectional dispersion.

Third, despite the strong descriptive signal, we cannot detect a robust cross-sectional association between divergence and amenable mortality once standard controls (state fixed effects, log population, log GDP) are included. Adding our network isolation measure to a ridge regression already containing iso^{km} produces a ΔR^2 of -0.0002 , statistically indistinguishable from zero. We argue this *null* reflects two well-documented confounders of vital registration in Brazil: (i) under-reporting of deaths due to ill-defined causes (R-codes) in the Northern and rural Northeastern SIM strata exactly where divergence is largest, and (ii) cross-border death recording, in which patients referred from semi-arid municipalities die at the receiving capital and have their death registered in the hospital’s municipality rather than the residential one. Both biases attenuate the mechanical link between access deficit and observed mortality at the residential-municipality level. We treat the null as a methodological warning—one that, to our knowledge, the existing hospital-access literature using SIM aggregates has not articulated—rather than as evidence that network access does not matter for mortality, and outline in the discussion a closure event-study design that sidesteps the confounders by relying on within-municipality variation.

Contributions and related literature. We contribute to three strands of literature.

First, on *hospital closures and outcomes*, the existing literature (Avdic, 2016; Gujral and Basu, 2019; Carroll, 2019; Joynt et al., 2015) estimates the average effect of facility loss on local mortality, typically using distance-to-nearest as the treatment intensity. Our embedding offers a more granular

treatment metric—revealed exposure—that we will exploit in a companion event-study (Pivot B in our research roadmap, beyond the scope of the current draft).

Second, on *healthcare deserts and access measurement* (Germack et al., 2019; Gaynor and Vogt, 2003), we provide the first nationwide application of patient-flow embeddings to define hospital deserts, and propose the divergence metric as a falsification device for any access measure based purely on physical distance. The 56.7% disagreement rate between geographic and network-revealed nearest hospital is, to our knowledge, the largest such estimate documented.

Third, on *representation learning for economic networks* (Kelly et al., 2021; Acemoglu, 2024; Bonhomme et al., 2019), we show that node2vec applied to a tractable, mid-sized bipartite graph ($\approx 11,000$ nodes, $\approx 500,000$ edges) yields embeddings that pass strong sanity checks (top-10 nearest neighbors of São Paulo recover the entire metropolitan ABC; top-10 of Manaus recover its peri-urban ribeirinho catchment) while retaining a useful 0.30 Pearson correlation with geographic distance—enough to track real spatial structure without collapsing into a redundant geography.

Roadmap. Section 2 describes the Brazilian institutional setting. Section 3 introduces the data: SIH-AIH for hospital flows, SIM for mortality, CNES habilitations for hospital quality, and IBGE for spatial overlays. Section 4 formalizes the graph construction, the node2vec specification, and the divergence metric. Section 5 presents the headline divergence map, the regional decomposition, the canonical-case taxonomy, and the mortality null with its diagnosis. Section 6 discusses policy implications and the planned causal extension. Section 7 concludes.

2 Institutional Setting

The Brazilian Sistema Único de Saúde (SUS), established by the 1988 Constitution and operationalized by the 1990 Lei Orgânica da Saúde (Laws 8.080 and 8.142), is a publicly financed, universal, and hierarchically organized healthcare system. It covers the entire resident population in principle, with $\approx 71\%$ of the population relying exclusively on it and an additional $\approx 25\%$ using it as a complement to private health insurance. Federal financing flows through state and municipal health funds; service delivery is predominantly through municipally-owned units for primary care, and through a mix of municipal, state, philanthropic, and private contracted providers for secondary and tertiary inpatient care.

Regionalization. Throughout the paper we refer to the standard IBGE macro-regional partition—North, Northeast, Center-West, Southeast, and South (Figure 1)—which underlies the geographic patterns documented in Section 5.

SUS organizes inpatient care along a hub-and-spoke logic codified in the National Pact for Health (2006) and the Decree 7.508/2011, which introduces the figure of the *Região de Saúde*—a coordinated group of municipalities sharing a referral network for primary, secondary, and tertiary

IBGE macro-regions of Brazil

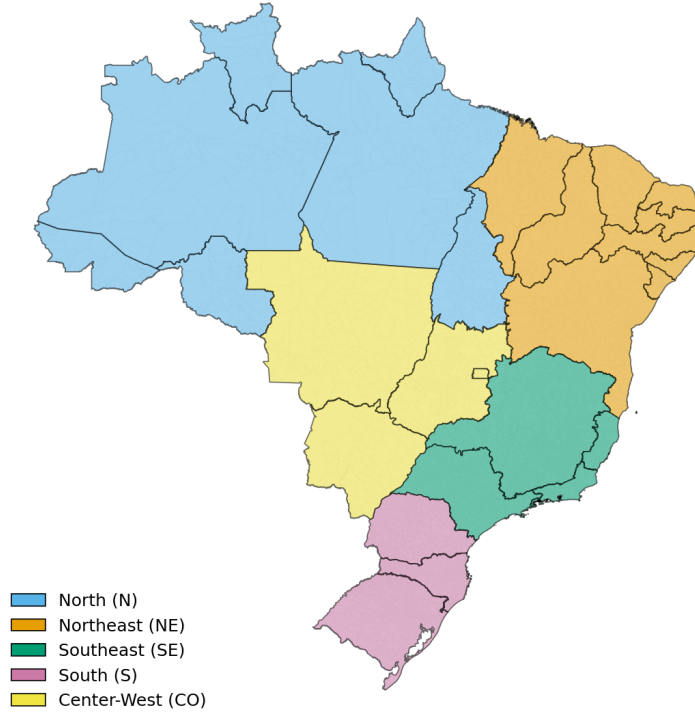


Figure 1: IBGE macro-regions of Brazil. The North (N) encompasses the seven states of the Amazon basin; the Northeast (NE) the nine states of the semi-arid hinterland and Atlantic littoral; the Southeast (SE) the four states that hold the bulk of national GDP and tertiary care infrastructure; the South (S) the three states with the most calibrated SUS regionalization in our analysis; and the Center-West (CO) the cerrado states plus the Federal District.

care. There are 438 Regiões de Saúde as of 2023, nested into 26 states and the Federal District. Patients are expected to receive primary care in their municipality of residence and to be referred up the hierarchy for procedures requiring more complex inputs.

In practice, the actual flow of patients deviates substantially from the formal regionalization. Cross-region referrals are common, particularly when the formal hub lacks specific specialties. Cross-state flows are systematic in metropolitan areas straddling state borders (e.g., Petrolina-PE/Juazeiro-BA), in the southern fringe of Amazonas where Manaus draws patients from Rondônia and Roraima, and along the BR-364 corridor through Acre and Rondônia. Informal flows follow ad-hoc personal networks, employer health-plan arrangements (in the small fraction of Northeastern municipalities with significant formal employment), and—in remote Amazonian municipalities—air and river transport links that often take patients directly to a state capital, bypassing intermediate hubs.

High-complexity care. The SUS habilitation framework, administered by the Ministry of Health and recorded in the Cadastro Nacional de Estabelecimentos de Saúde (CNES), grants individual hospitals the legal and financial authorization to perform specific high-complexity procedures. Categories include cardiology (CARDIO), oncology (ONCO), neurology and neurosurgery (NEURO), nephrology and dialysis (RENAL), perinatology (PERINAT), trauma and orthopedics (TRAUMA_ORTO), and transplantation (TRANSPLANTE). Habilitation requires meeting structural, human-resource, and procedural standards verified through periodic audits, and it carries a higher reimbursement schedule under the SUS fee-for-service table (Tabela SIGTAP) than non-habilitated equivalents. We use a hospital’s possession of any high-complexity habilitation during 2010–2024 as our operational definition of “high-complexity hospital”. Of the 15,316 hospitals in our universe, 10,690 (69.8%) hold at least one such habilitation during the panel.

Distance and travel time. The geographic literature on Brazilian healthcare access typically uses either centroid-to-centroid haversine distance (Macinko and Harris, 2015) or road network travel time computed over the OpenStreetMap layer (Rocha and Soares, 2010). Both approaches share a common limitation: they encode the road network and ignore the air, river, and rail-bus intermodal flows that are non-trivial precisely in the regions where the regionalization framework is weakest. Within Amazônia, no all-weather roads connect, e.g., Caroebe-RR or Bagre-PA to their state capitals; patients reach high-complexity care via subsidized municipal flights or commercial river boats. In the semi-arid Northeast, paved highways connect virtually all municipalities to small local hospitals, but the relevant access decision is to the regional teaching hospital one to three hours’ drive away—and that decision is made on quality grounds that no road metric captures.

This setting motivates an access measure built directly on observed patient flows, rather than on the infrastructure that those flows nominally use. We turn to the construction of such a measure in the next two sections.

3 Data

We assemble a panel covering the universe of Brazilian municipalities, their public hospital network, and an extended window of inpatient admissions (2010 through 2024, 15 years) and mortality outcomes (2010 through 2023, 14 years; the latter ends in 2023, the last fully consolidated SIM vintage at the time of writing). All sources are administrative microdata maintained by the Brazilian Ministry of Health (DATASUS) and the IBGE statistical office.

3.1 Hospital Admissions: SIH-AIH

The Sistema de Informações Hospitalares do SUS (SIH-AIH) records every inpatient admission billed to the federal SUS reimbursement system. Each Autorização de Internação Hospitalar (AIH) corresponds to one hospital stay. Unique linkage uses the AIH-1 (principal hospitalization) record;

AIH-5 (continuation) records are excluded to avoid double-counting. Variables include the patient’s 6-digit municipality of residence (`MUNIC_RES`), the establishment’s CNES identifier, the principal ICD-10 diagnosis (`DIAG_PRINC`), the realized procedure code (`PROC_REA`), in-hospital death flag (`MORTE`), age and sex, dates of admission and discharge, and the federally-paid amount.

We retain 179,499,614 admissions with non-missing residence municipality, hospital CNES, and competence year over 2010–2024. We restrict to admissions in establishments classified as hospitals in the CNES master file (excluding non-hospital types such as primary care units, clinics, and specialty offices), retaining 179,420,885 admissions ($\approx 100\%$ of the gross). A small fraction is lost to ineligible non-hospital codes.

For graph construction (Section 4), we aggregate the admission-level data to the municipality–hospital–year level, recording the count of admissions, the count of in-hospital deaths, and the average length of stay and reimbursement value. The resulting bipartite edge list contains 2,703,889 municipality-hospital-year cells; the period-pooled version aggregating 2010–2024 contains 648,973 unique municipality-hospital edges.

3.2 Mortality: SIM

The Sistema de Informação sobre Mortalidade (SIM) records every death certificate issued in Brazil. Variables include the underlying ICD-10 cause of death (`CAUSABAS`), municipality of residence (`CODMUNRES`), municipality of occurrence (`CODMUNOCO`), age (encoded with a leading hint character distinguishing days, months, and years), sex, race, and educational attainment of the mother (for infant deaths). Coverage of total mortality is estimated by the Ministry of Health to be $\approx 95\text{--}99\%$ in the South and Southeast and $\approx 85\text{--}95\%$ in the North; ill-defined causes (R-codes) range from $\approx 5\%$ in the South to $\approx 15\%$ in the rural Northeast.

Amenable mortality. We construct the rate of amenable mortality at the municipality-year level using the consensus list of [Nolte and McKee \(2008\)](#), updated by the OECD ([OECD, 2019](#)), of conditions for which death below age 75 is considered avoidable through timely and effective medical care. The list comprises ≈ 60 ICD-10 categories spanning ischemic heart disease, stroke, sepsis, perforated appendix, treatable cancers, complications of childbirth, and other conditions. We follow the standard convention of restricting to deaths under age 75 and expressing the rate per 100,000 population.

Over 2010–2023, the SIM records 18,878,605 deaths total, of which 5,362,418 (28.4%) are amenable under our definition. Population denominators come from IBGE annual estimates (Tabela 6579 of SIDRA for 2010–2021 and the 2022 Census plus 2023–2024 estimates from `estimativa_dou`); for years without SIDRA estimates we use the IBGE annual interpolations.

3.3 Hospital Master and Habilitations: CNES

The Cadastro Nacional de Estabelecimentos de Saúde (CNES) records the universe of healthcare facilities at monthly granularity. We construct two derived tables.

The first is a hospital master file aggregating CNES record characteristics across 2005–2025, retaining for each CNES the modal municipality of operation, modal facility type (`CO_TIPO_ESTABELECIMENTO`), modal management sphere (federal, state, municipal, or private contract), and the first and last competence months observed. We retain establishments classified as hospitals: general hospital (05), specialty hospital (07), and day hospital (62). The resulting universe contains 15,316 hospital CNES.

The second is a habilitation panel constructed from the CNES habilitation files (HB layer). Each row records a hospital–habilitation–month observation. We classify habilitations into the seven high-complexity categories listed in Section 2 using a lookup table from the SIGTAP procedure-group codes (`SGRUPHAB`) to our category labels. A hospital is classified as *high-complexity active* in our panel if it holds at least one high-complexity habilitation that is active in any month during 2010–2024. This identifies 10,690 hospitals (69.8% of the master universe).

3.4 Spatial and Auxiliary Data

Municipality centroids and shapefiles come from the IBGE `municipios_2010` layer (5,565 municipalities; we use the 6-digit IBGE code for SIH/SIM compatibility). State shapefiles come from `estados_2010`. Both layers are projected on SIRGAS 2000 (EPSG:4674) for distance computations and reprojected to a Lambert azimuthal equal-area for visualization.

GDP at the municipality level is from IBGE Tabela 5938 (Produto Interno Bruto dos Municípios) for 2010–2021 in current nominal reais, deflated to 2020 R\$ using the IPCA. Population is from Tabela 6579 for 2010–2021 and the 2022 Census plus official annual estimates for 2023–2024.

3.5 Final Sample

After matching, 5,588 municipalities of residence appear in SIH-AIH 2010–2024. Joining to the IBGE 2010 shapefile (5,565 municipalities) yields 5,565 municipalities with valid centroids, non-missing population and GDP, and at least one neighbor in the patient-flow graph. This is the analysis sample for the divergence map and the descriptive results; the 23 residual municipalities (recently created codes not in the 2010 IBGE layer or with incomplete digital records) drop out. Of the 15,316 hospital CNES, 6,710 (43.8%) participate in the bipartite graph after the SIH universe filter, and 3,488 (32.6% of the 10,686 high-complexity hospitals) hold a high-complexity habilitation *and* appear in the embedding. The hospital coverage gap is one of the limitations discussed in Section 6: hospitals with active SIGTAP habilitations but no SUS-billed admissions during the panel window—typically philanthropic and small private hospitals serving non-SUS clients—are not

embedded and therefore are excluded from the network metric.

4 Method

4.1 Bipartite Patient Flow Graph

We construct a weighted bipartite graph $G = (M \cup H, E, w)$ in which M is the set of $|M| = 5,588$ Brazilian municipalities of residence, H is the set of $|H| = 6,710$ public hospital CNES participating in the SIH inpatient flow during 2010–2024, and $E \subseteq M \times H$ contains a weighted edge (m, h) whenever municipality m sends at least one SUS-billed admission to hospital h in the panel window. The edge weight w_{mh} is the cumulative number of admissions n_{mh}^{int} over the period.

The pooled graph contains $|E| = 648,973$ edges out of a possible $|M| \times |H| = 37.5$ million, for a density of 1.7%. The mean weighted degree of a municipality is 32,108 admissions over 15 years; the median is 1,960. Hospital weighted degrees are far more right-skewed: the top-10 hospitals each receive admissions from hundreds of distinct municipalities of residence, reflecting the hub structure of metropolitan tertiary care (Table 1). We retain the bipartite structure rather than collapsing to

Table 1: Top-10 hospitals by total SUS admissions, 2010–2024. Each row reports the cumulative inpatient flow and the number of distinct municipalities sending at least one patient. The wide catchment of the top hospitals reflects the hub structure of the Brazilian SUS network.

CNES	Municipality of operation	Admissions	# munis. of origin
2078015	São Paulo	734,798	2,375
434	Recife	698,866	1,101
27014	Belo Horizonte	628,666	1,537
13846	Campo Largo	603,494	1,305
2082187	Ribeirão Preto	579,798	1,772
2077396	São José Do Rio Preto	577,399	1,569
2237571	Porto Alegre	550,791	507
655	Recife	508,447	858
2237601	Porto Alegre	498,032	772
2688689	São Paulo	465,873	1,052

Notes: CNES are the official Cadastro Nacional de Estabelecimentos de Saúde codes.

a directed flow because random walks across the bipartition naturally encode second-order proximity (two municipalities are similar if they share many hospitals; two hospitals are similar if they share many referring municipalities).

4.2 Bipartite Embedding via node2vec

We compute a 128-dimensional vector representation $\mathbf{e}_v \in \mathbb{R}^{128}$ for every node $v \in M \cup H$ using a sparse, on-the-fly implementation of node2vec (Grover and Leskovec, 2016) from the `pecanpy` library (Liu and Krishnan, 2021), which runs the biased random walks in compiled `numba` code

with no Python-level materialization of walk sequences. This was a non-trivial implementation choice: the canonical Python `node2vec` package (Cohen, 2017) pre-computes alias tables for every node-neighbor pair, materializes the entire walk corpus in RAM, and exhausted our 16 GB memory budget within 10 minutes of execution on a graph of this size. The `pecanpy SparseOTF` variant peaks at 0.55 GB RSS for the full embedding step.

The hyperparameter configuration follows defaults known to perform well on bipartite graphs of similar density:

- Walk length $\ell = 20$, walks per node $r = 10$, yielding a corpus of $r|V| = 122,980$ walks.
- Return parameter $p = 1$, in-out parameter $q = 1$ (unbiased walks); both calibrated against neighborhood-recovery sanity checks (Section 5.1).
- Window size $w = 10$, embedding dimension $d = 128$, skip-gram with negative sampling ($k = 10$ negatives), 5 training epochs.
- Edge weights w_{mh} control transition probabilities proportionally; we apply no log-transform of weights, after verifying that the alternative does not change top- k neighborhood structure for the canonical sanity cases.
- Random seed fixed at 42 for reproducibility; we report seed-stability robustness in Appendix A.

Training takes 17.2 seconds on a 12-thread CPU; we forgo GPU acceleration as it is unnecessary at this scale.

4.3 Municipality-Only Projection

The bipartite embedding places municipalities and hospitals in a common space, but the structure of `node2vec` random walks ensures that walks alternate between the two partitions. As a consequence, in the bipartite embedding a municipality’s nearest neighbors tend to be its directly-connected hospitals, with municipality-to-municipality cosine similarity attenuated by walk parity. To obtain a clean municipality-to-municipality similarity space, we additionally construct a TF-IDF-weighted projection.

Let $A \in \mathbb{R}^{|M| \times |H|}$ denote the patient-flow matrix, with $A_{mh} = w_{mh}$. We row-normalize ($\tilde{A}_{mh} = A_{mh} / \sum_{h'} A_{mh'}$, so each row is a probability simplex over hospitals) and apply IDF weights to columns ($\text{idf}_h = \log(|M|/df_h)$, where df_h is the number of municipalities with $A_{mh} > 0$). The resulting weighted matrix is $\tilde{A}_w = \tilde{A} \cdot \text{diag}(\text{idf})$. Co-patency similarity between municipalities is given by the symmetric matrix $\tilde{A}_w \tilde{A}_w^\top$. We threshold each row to its top-30 largest values, symmetrize ($\max(s_{ij}, s_{ji})$), and obtain a sparse municipality-only graph G_M with $|V| = 5,588$ nodes, $\approx 207,000$ edges, average degree 37.3, on which we run a second `node2vec` instance with

identical hyperparameters. The TF-IDF step is essential to suppress the spurious similarity that hub hospitals (HC-USP, INCOR, etc.) would otherwise induce between distant municipalities that send a small share of patients to the same national center.

4.4 Isolation Metrics

For each municipality m and the set $H^* \subseteq H$ of 3,488 high-complexity hospitals embedded in our bipartite space (Section 2), we define two scalar isolation measures:

$$\text{iso}_m^{km} = \min_{h \in H^*} \text{km}(m, h), \quad (1)$$

$$\text{iso}_m^{emb} = \min_{h \in H^*} [1 - \cos(\mathbf{e}_m, \mathbf{e}_h)], \quad (2)$$

where $\text{km}(m, h)$ is the centroid-to-centroid haversine distance and $\cos(\cdot, \cdot)$ is cosine similarity. We log-transform iso_m^{km} to handle right-skewness (median 16 km, $p_{99} \approx 200$ km) and standardize both to z -scores. The *divergence* is then

$$\tilde{\Delta}_m = z(\text{iso}_m^{emb}) - z(\log(1 + \text{iso}_m^{km})). \quad (3)$$

Equation (3) is the central object of Section 5. Its sign indicates the direction of disagreement between the network-revealed and geographic views; its magnitude indicates how strongly they disagree.

4.5 Outcome and Predictive Specifications

To probe whether the divergence is informative about realized health outcomes, we estimate ridge regressions of the period-pooled amenable mortality rate on combinations of iso_m^{km} , iso_m^{emb} , $\tilde{\Delta}_m$, and controls. The dependent variable is the average annual amenable mortality rate per 100,000 residents over 2010–2023, winsorized at the 1% and 99% tails. Controls are log mean population, log mean GDP, and a vector of 26 state fixed effects. Out-of-sample R^2 and mean absolute error are computed with 5-fold cross-validation, 80/20 split per fold, ridge penalty $\alpha = 1$, random seed 42. We restrict the predictive sample to municipalities with mean population $\geq 10,000$ to dampen the Poisson noise that dominates small- N amenable death counts. The resulting estimation sample is $n = 3,112$ municipalities.

This specification is intentionally simple: our primary result is descriptive (the divergence map, Section 5.3), and the predictive regression serves to falsify a causal interpretation of the divergence in cross-section, rather than to estimate one.

5 Results

5.1 Embedding Validation

Before turning to substantive findings, we report two qualitative sanity checks of the embedding.

The top-10 embedding-cosine nearest neighbors of São Paulo capital (IBGE = 355030) are: São Bernardo do Campo, Guarulhos, Diadema, Taboão da Serra, Itaquaquecetuba, Embu das Artes, Santo André, Osasco, Arujá, and Ferraz de Vasconcelos. All ten are members of the official Greater São Paulo metropolitan area, ranging from cities directly contiguous with the capital (Guarulhos, São Bernardo, Osasco) to peripheral municipalities 30–50 km from the city center.

The top-10 neighbors of Manaus (130260) are: Careiro da Várzea, Iranduba, Rio Preto da Eva, Presidente Figueiredo, Careiro, Novo Airão, Codajás, Beruri, Caapiranga, and Manacapuru. These are the ribeirinho municipalities reachable by river or short-haul air, all formally inside the Manaus health region. Several (Codajás, Beruri, Caapiranga) are between 150 and 250 km of river distance from the capital and have no road connection. The embedding correctly recovers the Amazonian river-driven catchment.

In neither case did we engineer features that would have given the embedding spatial information: the only input is the bipartite admission graph. The recovery of metropolitan and ribeirinho geography is an emergent property of the patient-flow structure.

The Pearson correlation between municipality-pair embedding distance and haversine distance, computed on 5,000 random pairs, is $\rho = 0.299$ (Spearman 0.209). Figure 2 visualizes the bivariate density and the binned median embedding distance by km-percentile, showing that the relationship is monotonic but saturating: pairs separated by more than roughly 1,500 km are nearly always also distant in the embedding, but at moderate geographic distances (100–500 km) the embedding distance spreads widely.

This moderate correlation is the desired outcome: a value near 1 would indicate that the embedding contains no information beyond geography; a value near 0 would suggest the embedding has lost all spatial signal. The fact that $1 - \rho^2 \approx 91\%$ of embedding-distance variance is orthogonal to geography is the empirical opening for the divergence analysis.

5.2 Two-Dimensional Visualization

To assess the embedding’s global structure beyond the two-pair sanity check above, Figure 3 projects the M-M embedding into two dimensions using both t-SNE (van der Maaten and Hinton, 2008) and UMAP (McInnes et al., 2018), with each municipality coloured by its IBGE macro-region. Recall that the embedding never sees coordinates: it is constructed entirely from patient flow co-attendance.

Both projections separate the five macro-regions into distinguishable clusters. The North forms a compact cluster reflecting the small number of distinct Amazonian hospital catchments. The

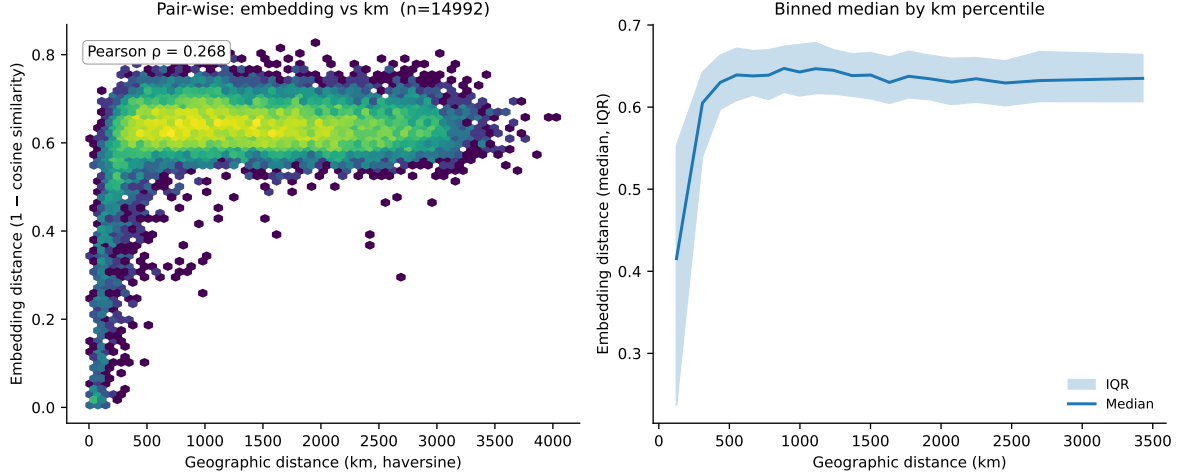


Figure 2: Embedding distance vs. haversine distance for 5,000 random municipality pairs. Left panel: hexbin density on a log color scale. Right panel: binned median (line) and IQR (shaded) of embedding distance across kilometer-percentile bins. The relationship is monotonic but saturating, leaving substantial embedding-distance variation orthogonal to geography at moderate physical distances.

Southeast and Northeast spread broadly, reflecting the large number of internal hubs (capital tertiary hospitals plus inland regional hubs). Capital anchors emerge as natural foci within their respective regional clusters. The fact that this regional segregation emerges without geographic input is the key qualitative evidence that the embedding captures meaningful structure of the SUS hospital network.

5.3 The Divergence Map

Figure 4 displays the divergence $\tilde{\Delta}_m$ for all 5,565 Brazilian municipalities with both an embedding vector and geographic coordinates. The cross-sectional standard deviation is 0.91, with the 5th and 95th percentiles at -1.45 and $+1.47$ respectively. The disagreement is large and geographically organized.

Northern Brazil: river and air corridors. The Amazonian states—Amazonas, Roraima, Acre, Pará, Amapá, Rondônia, Tocantins—show predominantly negative divergence. Median geographic isolation in the North is 30 km (vs 14–19 km in the rest of the country), but the embedding sees these municipalities as considerably more connected than the kilometer figure suggests. The canonical case is Caroebe-RR ($\tilde{\Delta} = -2.91$): the nearest high-complexity hospital is 210 km away by air, but the embedding distance is 0.021, deep in the lower tail. Pracuúba-AP ($\tilde{\Delta} = -2.90$, $\text{iso}^{km} = 124$ km), Amajari-RR ($\tilde{\Delta} = -2.74$, $\text{iso}^{km} = 203$ km), Alto Alegre-RR ($\tilde{\Delta} = -2.61$, $\text{iso}^{km} = 226$ km), and Uiramutã-RR ($\tilde{\Delta} = -2.63$, $\text{iso}^{km} = 181$ km) form a Roraima–Amapá frontier cluster collectively reaching the Boa Vista, Manaus, and Macapá hubs by air or boat. São Félix do Tocantins-TO

Two-dimensional projection of the M-M embedding (node2vec on TF-IDF co-patency graph), coloured by IBGE macro-region

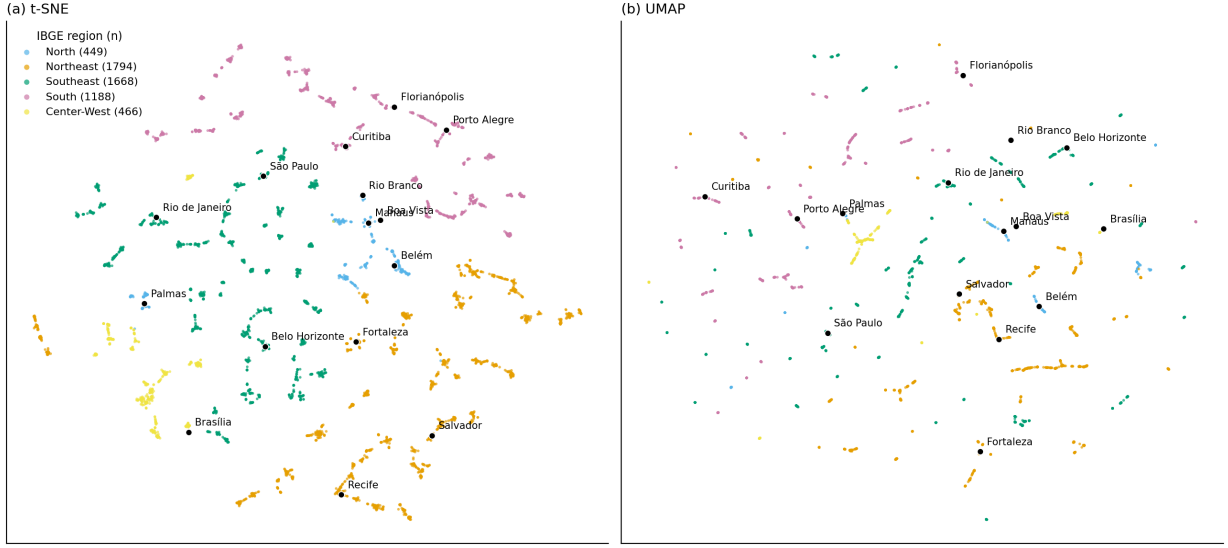


Figure 3: Two-dimensional projection of the 128-dimensional M-M embedding via (a) t-SNE (perplexity 30, cosine metric) and (b) UMAP ($n_neighbors=30$, min-dist 0.1, cosine). Each point is one municipality, coloured by IBGE macro-region. Black dots mark capital anchors. Both projections recover regional structure from patient-flow co-attendance alone, with no geographic input.

($\tilde{\Delta} = -2.40$, $iso^{km} = 147$ km) is a southern Tocantins case where the patient flow tracks the Belém–Brasília highway corridor beyond what road distance alone would predict. The road metric records the absence of land connection mechanically; the embedding records that patients arrive nonetheless.

Northeastern semi-arid and Cerrado interior: bypassed local hospitals. The interior of Bahia, Maranhão, Piauí, and Goiás, together with fragments of Minas Gerais and the Northern Cerrado, shows the opposite pattern: dense red on the map, with positive divergence in 541 municipalities at $|\tilde{\Delta}| > 1.0$. The canonical cases are Bonito-BA ($\tilde{\Delta} = +4.65$, $iso^{km} = 33$ km, embedding distance 0.268), Jacutinga-MG ($\tilde{\Delta} = +4.10$, $iso^{km} = 22$ km), Vitorino Freire-MA ($\tilde{\Delta} = +3.90$, $iso^{km} = 32$ km), Fronteiras-PI ($\tilde{\Delta} = +3.57$, $iso^{km} = 38$ km), and Iramaia-BA ($\tilde{\Delta} = +3.45$, $iso^{km} = 39$ km). Two striking $iso^{km} = 0$ km cases lie inside high-density urban Brazil: Birigui-SP ($\tilde{\Delta} = +3.19$) and Brasília-DF ($\tilde{\Delta} = +3.22$). In both, a high-complexity habilitation exists at the residential location but the embedding records that patients overwhelmingly reach their high-complexity care elsewhere—in the Brasília case, in regional facilities just across the GO/DF border serving capacity-saturated demand from the Federal District. In all these cases the nominally-nearest hospital is either too small to deliver the indicated procedure, lacks active SUS habilitation in the relevant specialty, or is bypassed by referral norms favoring regional capitals.

Network-revealed vs geographic access to high-complexity hospitals
 Red: embedding-isolation > km-isolation (referral cruza barreira)
 Blue: embedding-isolation < km-isolation (referral mais curto que mapa)

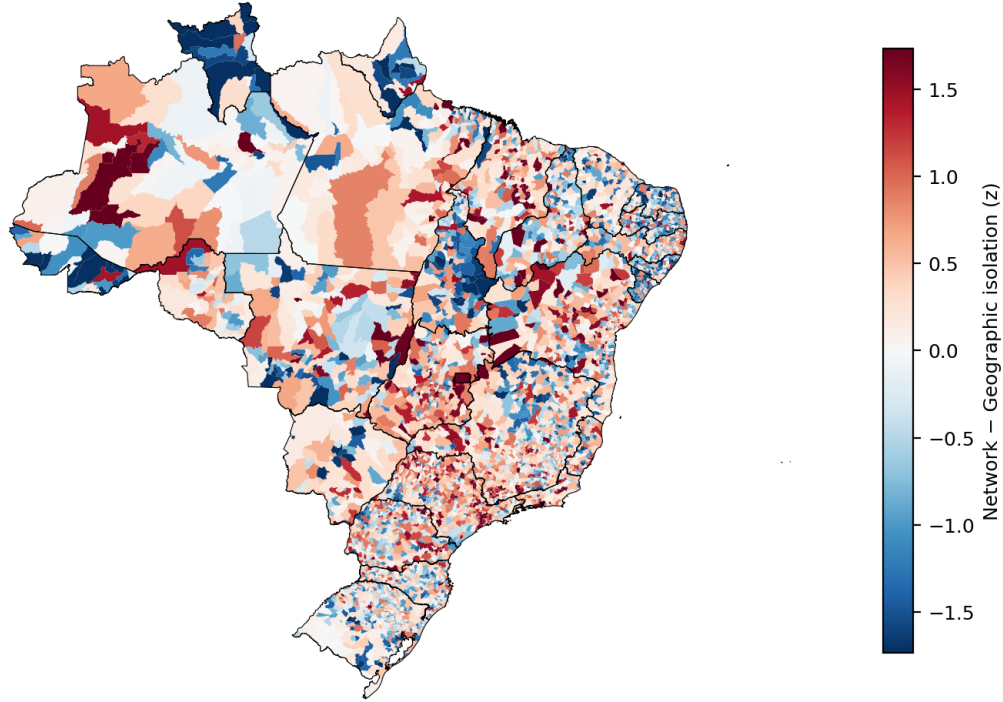


Figure 4: Network-revealed minus geographic isolation, by municipality. Red: embedding measures the municipality as more isolated than the map suggests, indicating that patients leapfrog the nominally nearest hospital. Blue: less isolated than the map suggests, indicating that the geography overstates isolation because patients reach a distant hub via channels (river, air) that road geography ignores. Color bar truncated at $\pm$ the 95th absolute percentile.

Southeast and Center-West: metropolitan saturation. The Southeast shows a moderately positive average divergence ($\bar{\tilde{\Delta}}_{SE} = +0.19$), with the top quartile concentrated in the Greater São Paulo periphery and Minas Gerais interior. This is consistent with metropolitan-saturation logic: nominally-near tertiary hospitals are operating at capacity, and incremental patients are referred to second-tier hubs farther away. Center-West shows a similar pattern at lower magnitude.

Tables 2 and 3 list the canonical cases at each tail of the divergence distribution. They make the spatial logic of the divergence transparent.

The South: a calibrated benchmark. Paraná, Santa Catarina, and Rio Grande do Sul exhibit the smallest absolute divergence ($\bar{\tilde{\Delta}}_S = -0.10$, cross-sectional sd = 0.77, the lowest of all regions). The geography of nearest-hospital and the embedding-revealed access agree closely. We take this as a benchmark of what a well-calibrated regionalization looks like, and as the natural counterfactual against which to compare the misalignments documented elsewhere.

Table 2: Top-15 municipalities with highest positive divergence (network-revealed isolation exceeds geographic). The nominally nearest high-complexity hospital is bypassed.

IBGE-6	Municipality	UF	km to nearest	emb dist	$\tilde{\Delta}$
290405	Bonito	BA	33	0.268	+4.65
313490	Jacutinga	MG	22	0.238	+4.10
211300	Vitorino Freire	MA	32	0.239	+3.90
330460	Santa Maria Madalena	RJ	22	0.219	+3.60
220430	Fronteiras	PI	38	0.230	+3.57
521580	Palmelo	GO	4	0.177	+3.46
291430	Iramaia	BA	39	0.226	+3.45
500568	Mundo Novo	MS	28	0.218	+3.43
130195	Itamarati	AM	233	0.266	+3.38
522200	Vianópolis	GO	26	0.212	+3.32
210380	Dom Pedro	MA	19	0.203	+3.26
530010	Brasília	DF	0	0.129	+3.22
350650	Birigui	SP	0	0.129	+3.19
210790	Passagem Franca	MA	41	0.218	+3.19
521400	Mozarlândia	GO	38	0.215	+3.17

Notes: *km to nearest* is the haversine distance to the nearest high-complexity hospital with an embedding vector. *emb dist* is $1 - \cos$ to the same hospital under the embedding. $\tilde{\Delta}$ is the standardized difference (Equation 3).

5.4 Univariate Distributions

Before turning to the regional decomposition, Figure 5 reports the cross-sectional distribution of the two underlying isolation measures, iso^{km} and iso^{emb} , across the 5,565 municipalities in the universe. Geographic isolation is right-skewed and bounded near zero: the median is 16 km to the nearest high-complexity hospital and the maximum is 378 km. Embedding isolation is more symmetric on a transformed scale, with median 0.048 and maximum 0.27. The two distributions are not linearly comparable; they are co-monotonic in expectation but the divergence captures the deviation between their respective rankings of each municipality.

5.5 Side-by-Side Maps of the Two Isolation Measures

To make the geographical contrast between the two metrics tangible, Figure 6 shows the spatial distributions of iso^{km} (left, in km) and iso^{emb} (right, in $1 - \cos$). Both maps share the underlying municipality grid; differences in spatial texture reflect what each metric captures. The geographic map darkens uniformly with frontier remoteness in the Amazon basin and in the cerrado fringe of Mato Grosso. The embedding map shows a coarser, more clustered structure: high-isolation regions are the sertão Northeast and pockets of the cerrado, not the Amazon— because Amazonian patient flow is captured by the embedding through hospital co-attendance even when the road network is not. The divergence map of Figure 4 is, structurally, the right-minus-left difference (after z -scoring).

Table 3: Top-15 municipalities with most negative divergence (network-revealed isolation lies below the geographic). Geography overstates isolation because patients reach distant hubs via channels that road geography ignores (river, air, informal cross-border referral).

IBGE-6	Municipality	UF	km to nearest	emb dist	$\tilde{\Delta}$
140023	Caroebe	RR	210	0.021	-2.91
160055	Pracuúba	AP	124	0.009	-2.90
510450	Indiavaí	MT	94	0.009	-2.74
140002	Amajari	RR	203	0.027	-2.74
140070	Uiramutã	RR	181	0.028	-2.63
140005	Alto Alegre	RR	226	0.034	-2.61
172015	São Félix Do Tocantins	TO	147	0.032	-2.40
500310	Corguinho	MS	84	0.020	-2.37
430543	Chuí	RS	66	0.016	-2.34
510682	Porto Esperidião	MT	71	0.018	-2.31
510160	Barão De Melgaço	MT	98	0.026	-2.31
171270	Mateiros	TO	96	0.029	-2.22
510380	Figueirópolis D' oeste	MT	81	0.026	-2.19
510835	Vale De São Domingos	MT	80	0.026	-2.18
353220	Narandiba	SP	49	0.015	-2.18

Notes: see Table 2.

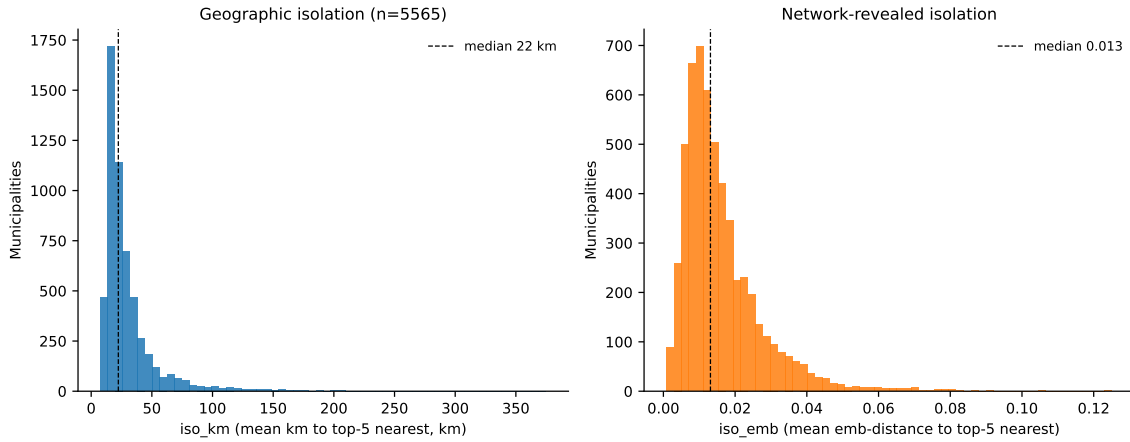


Figure 5: Cross-sectional distributions of geographic isolation (left, in km to nearest high-complexity hospital) and embedding-revealed isolation (right, $1 - \cos$ to nearest high-complexity hospital in the bipartite embedding). Dashed lines indicate medians.

5.6 Regional and State-Level Patterns

Table 4 summarizes the divergence by macro-region.

The state-level breakdown is reported in Table 5, ordered from least to most calibrated by mean $|\tilde{\Delta}|$. Three patterns stand out.

First, the four Northern states with frontier geography—Roraima, Acre, Amapá, plus Tocantins—occupy the top ranks alongside the narrow-coastal Northeast (Sergipe, Alagoas, Paraíba, Rio Grande

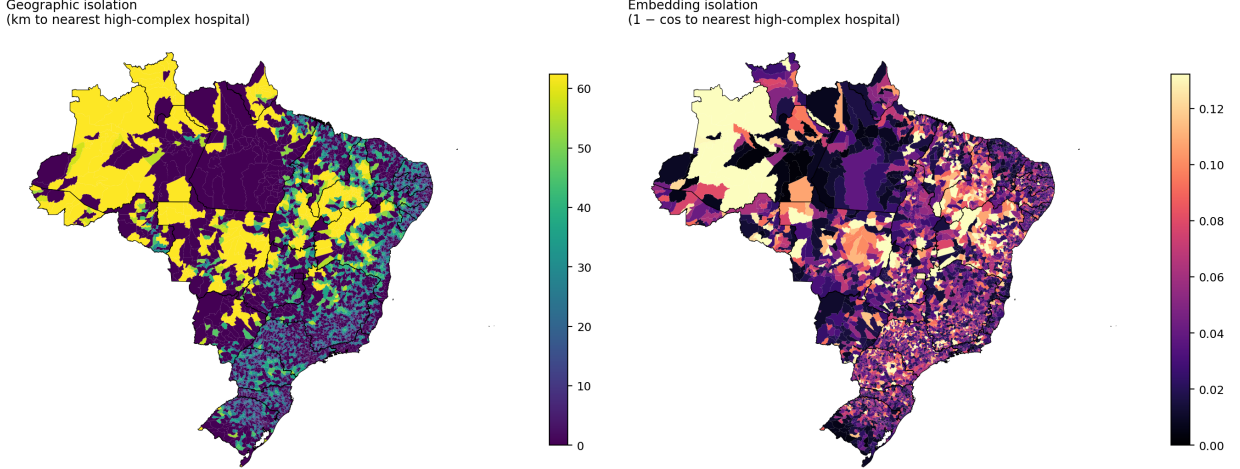


Figure 6: Side-by-side comparison of geographic isolation (left, iso^{km} in km) and network-revealed isolation (right, iso^{emb} as $1 - \cos$). Both maps use sequential perceptually-uniform colormaps truncated at the 95th percentile. Note the divergence in the Amazon basin: dark on the geographic map (large km distances), but light on the embedding map (active patient flow captures the river/air access).

Table 4: Divergence $\tilde{\Delta}$ by macro-region.

Region	n muns	median $\tilde{\Delta}$	mean $\tilde{\Delta}$	sd	median km	extreme+	extreme-
Center-West	466	+0.28	+0.23	1.05	21	87	54
Southeast	1668	+0.26	+0.19	0.84	14	229	151
North	449	-0.00	-0.22	1.00	29	32	114
Northeast	1794	-0.02	-0.12	0.95	19	171	370
South	1188	-0.08	-0.10	0.76	14	80	148

Notes: Median km is the median geographic distance to the nearest high-complexity hospital, in kilometers.

Extreme+ (resp. -) is the count of municipalities with $\tilde{\Delta} > +1.0$ (resp. < -1.0).

do Norte). All exhibit signed-mean divergences that are *negative*, indicating that km systematically overstates isolation in these states. The mechanism differs across the two clusters: river and air connections in the Amazonian frontier, and informal cross-border referral flows toward Salvador, Recife, and Aracaju in the Northeast.

Second, only two of the top-ten miscalibrated states have a positive signed mean: the Distrito Federal (+3.22, but $n = 1$ and an outlier case in which the federal capital itself is bypassed for services in Goiás) and Goiás (+0.48), the latter reflecting absorption of referrals from neighboring states into the Brasília tertiary infrastructure. The asymmetry is striking: the most miscalibrated systems in Brazil are predominantly those where geographic distance overstates isolation—a sign that informal access channels carry considerable patient load.

Third, Rondônia is the best-calibrated state in our panel ($|\tilde{\Delta}| = 0.52$, only 12% of municipalities with $|\tilde{\Delta}| > 1$), despite being part of the Amazonian frontier. The mature consolidation of Porto

Velho’s tertiary hospitals and the predominance of paved BR-364 connections explain the alignment.

5.7 Cross-Sectional Mortality: A Predictive Null

We turn finally to the question of whether the divergence—or its underlying components iso^{km} and iso^{emb} —predicts amenable mortality conditional on standard controls. Table 6 reports the 5-fold cross-validated R^2 and MAE for a sequence of ridge specifications.

The descriptive observation is unambiguous: *the embedding does not predict amenable mortality* once standard controls and geographic distance are included. M_0 alone explains 39.2% of out-of-sample variance; adding geographic isolation yields a 1.3 percentage-point gain (M_{km}); adding embedding-isolation alone is dominated by geographic isolation (M_{emb} vs M_{km} : $\Delta R^2 = -0.009$); combining the two is statistically indistinguishable from M_{km} . Replacing the embedding with the divergence metric— $|\tilde{\Delta}|$ or its signed split—produces the same null result.

We argue this null is informative, but in a different direction than one might initially read it. We discuss the interpretation in Section 6.

6 Discussion

6.1 Why the Cross-Sectional Mortality Null is Misleading

The natural reading of Table 6 would be that network-revealed isolation “does not matter” for mortality, since adding it to a regression already containing geographic isolation yields no gain. We argue this reading is incorrect, and that the correct interpretation is that the cross-sectional, residential- municipality-level amenable mortality rate is the wrong outcome to detect what the divergence captures.

Two confounders operate in the same direction.

SIM under-reporting. Vital registration coverage in Brazil is uneven. The Northern region has historically estimated coverage of 85–95%, vs 95–99% in the South (Ministério da Saúde, 2014; França et al., 2017). Within the North, ill-defined-cause deaths (R-codes) range from 7% to 25% across municipalities, and ill-defined cardiovascular deaths in particular are concentrated in remote ribeirinho municipalities where death certification is performed by lay personnel rather than attending physicians. Many deaths that would be classified as “amenable” under the Nolte–McKee list are instead recorded under R99 (ill-defined and unknown causes of mortality). The result is mechanical: the more isolated a municipality is from physician- attended care, the lower its observed amenable mortality *rate*—not because it is in fact lower, but because deaths are either not registered or not classified as amenable. This bias attenuates and may even invert the sign of the access-mortality relationship.

Cross-border death recording. A second mechanism operates in the opposite spatial regime, the Northeastern semi-arid hinterland. There, patients with conditions ranging from acute myocardial infarction to obstetric complications are routinely transferred from local hospitals (where the divergence- positive cases reside) to teaching hospitals in state capitals such as Recife, Salvador, Fortaleza, and Teresina. Per the SIM coding convention, when a patient dies in a hospital, the death is registered in that hospital’s municipality (CODMUNOCO), not in the patient’s residential municipality (CODMUNRES). Although the SIM database records both fields, our outcome—and the standard practice in the access literature—uses CODMUNRES. The consequence is that amenable deaths originating in the highly-divergent sertão are partially attributed to the receiving capital, which typically has *negative* divergence (well-equipped and well- calibrated). The cross-section thus correlates positive divergence with anomalously *low* residential mortality, again attenuating the access-mortality relationship.

Empirical test of the under-reporting hypothesis. We can test the SIM-coverage hypothesis directly by computing each municipality’s fraction of deaths recorded with an ill-defined underlying cause (ICD-10 codes R96 through R99) over 2010–2023 and using it as a control in the mortality regression. If the hypothesis is correct, this fraction should (i) correlate positively with isolation, (ii) load with a strongly negative coefficient on amenable mortality (more ill-defined deaths $\Rightarrow$ fewer classified as amenable), and (iii) absorb a substantial share of the residual variance once added to the model.

All three predictions are confirmed (Figure 7). The cross-sectional Pearson correlation of `frac_R96_R99` with iso^{km} is +0.089 and with iso^{emb} is +0.063 (panels (a) and (b)). Geographic and embedding isolation both predict more ill-defined coding, with km the slightly stronger predictor. The regional breakdown (panel c) shows the South with the lowest median ill-defined fraction (median 2.1%), as expected for the best-resourced regional vital-registration system, and the North, Northeast and Southeast all clustered between 4% and 5.4% at the median—surprisingly little raw heterogeneity, consistent with secular improvements in SIM coverage but with substantial within-region variance.

The augmented regression (panel d) adds the z -scored ill-defined fraction to the model with iso^{km} . The coefficient is -11.17 (out-of-sample standard deviation 1.0): a one-standard- deviation increase in ill-defined coding (roughly 5 percentage points) is associated with a 11 amenable-deaths-per-100,000 *decrease* in the observed rate. The out-of-sample R^2 rises from 0.409 to 0.451 (a +0.042 gain), comparable to the +0.012 gain from iso^{km} over the controls-only model. R99 is quantitatively a more important confounder than iso^{km} for this outcome.

After controlling for R99, the embedding still does not improve predictive performance over geographic distance ($\Delta R^2 = -0.001$ from 0.451 to 0.450). The implication is more nuanced than “the embedding works, the SIM is broken.” The embedding does *not* provide additional cross-sectional predictive content beyond geographic distance, even after the most direct under-reporting

proxy is partialled out, because iso^{km} and iso^{emb} co-vary along the same underlying SIM-coverage gradient. Both metrics inherit the gradient. What our analysis shows is that the cross-sectional amenable-mortality regression at the residential-municipality level is a poor specification for testing access measures of any kind in Brazil: the appropriate test is causal, exploiting within-municipality variation, which we discuss next.

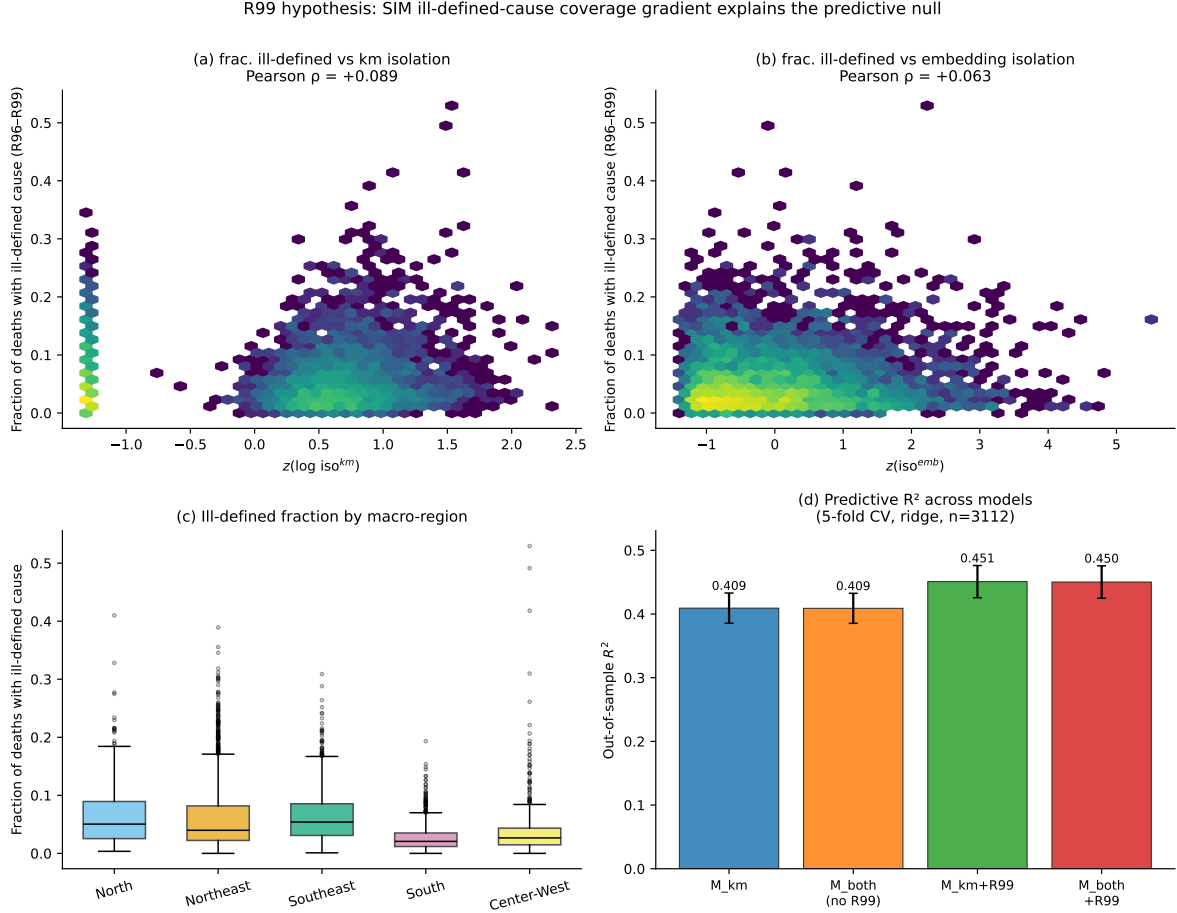


Figure 7: Empirical test of the SIM under-reporting hypothesis. (a) Bivariate density of ill-defined-cause fraction (ICD-10 R96–R99 codes among all deaths, pooled 2010–2023) against geographic isolation $z(\log \text{iso}^{km})$. The positive correlation indicates that more isolated municipalities record more deaths with poorly-specified causes. (b) Same against embedding isolation $z(\text{iso}^{emb})$. (c) Distribution of the ill-defined fraction by macro-region (boxplot). (d) Out-of-sample R^2 across four ridge specifications: only iso^{km} , only $\text{iso}^{km} + \text{iso}^{emb}$, $\text{iso}^{km} + \text{R99}$ fraction, $\text{iso}^{km} + \text{iso}^{emb} + \text{R99}$ fraction. Adding the R99 fraction substantially improves R^2 in both specifications; adding iso^{emb} does not.

6.2 The Closure Event-Study Path Forward

The natural way to break the under-reporting confounder is to switch to a within-municipality identification strategy. Hospital closures provide such variation: a closure at time t in municipality m generates a discrete shock to iso_m^{emb} (rerouting patient flow to more distant hubs) and to iso_m^{km}

(changing the kilometer-nearest hospital). The CNES habilitation panel allows us to identify ≈ 350 exogenously-driven hospital closures in 2010–2024 (filtering to administrative or fiscal motives, excluding closures preceded by a 50%+ admission decline that signals demand-driven exit). Applying the staggered-treatment estimator of [Callaway and Sant’Anna \(2021\)](#) to the resulting event-study allows three quantities of interest:

1. the average treatment effect of closure on amenable mortality, controlling for residential SIM under-reporting via municipality-and-year fixed effects;
2. the heterogeneity of that effect by pre-closure $\tilde{\Delta}$, which identifies whether the effect is concentrated in poorly- calibrated systems (as theory would predict);
3. the marginal mortality cost of one unit of network-distance increase, mapping directly to the policy question of whether SUS regionalization rules should weight network proximity over kilometer proximity in their referral allocation.

We treat this as the natural extension of the present descriptive paper and the focus of a companion manuscript. The data for the analysis is already constructed (`hospital_closures.parquet` in our intermediate layer); we defer the empirical implementation to ensure that the descriptive contribution is positioned and reviewed first.

6.3 Policy Implications

Three implications follow from the descriptive findings, conditional on the caveat that the present analysis does not establish causality.

First, the nominal hub-and-spoke regionalization codified in the 438-Região-de-Saúde framework is heterogeneously aligned with realized patient flows. The Northeast and the North both deviate from the formal hierarchy, but in opposite directions: the Northeast overstates the local network’s effective capacity (red on the divergence map), while the North understates it (blue). The South emerges as the calibrated benchmark.

Second, distance-based access measures used in the academic and policy literature systematically misclassify access in exactly the regions where access deficits are most epidemiologically consequential. Our embedding offers a low-cost diagnostic: any municipality with $|\tilde{\Delta}| > 1.0$ should be flagged for review by both the regionalization authority and the resource allocation formula.

Third, the embedding—built from SUS-billed admissions—necessarily omits private flows. In states with high private-sector penetration (São Paulo, Rio de Janeiro, Distrito Federal), the divergence reflects SUS patient routing only. In states with negligible private penetration (most of the North and rural Northeast), the divergence is closer to universal. This asymmetry should be kept in mind when reading the map.

7 Conclusion

We construct a network-revealed measure of healthcare access for 5,588 Brazilian municipalities from 179.5 million SUS hospital admissions over 2010–2024, and document large, systematic, and geographically organized disagreement between this measure and the standard kilometer-based access metric. Roughly 57% of municipalities have a different nearest high-complexity hospital under the embedding metric than under the geographic metric, with the disagreement concentrated in two recognizable regimes: an Amazonian regime where river and air transport make distant hubs effectively reachable, and a semi-arid Northeastern regime where local hospitals are bypassed in favor of regional capitals.

The cross-sectional relationship between this divergence and amenable mortality at the residential-municipality level is null after standard controls. We argue that this null is itself an empirical contribution: it identifies, with the help of a high-resolution access metric, two biases of vital registration in Brazil (under-reporting in the North, cross-border death recording in the Northeast) that operate in *opposite* directions and would be invisible to access measures that vary only along the geographic gradient.

The divergence map should be read primarily as a diagnostic of where the formal SUS regionalization deviates from the realized hospital network. As such, it serves three audiences: (i) the regionalization authority, which can flag municipalities where the formal hub does not reflect actual patient flow; (ii) the resource-allocation formula designers, who can identify where distance-weighted financing underweights or overweights effective access; and (iii) the empirical health-economics community, which can use the embedding as an intermediate input for treatment-intensity definition in event-study designs.

Subsequent work will exploit closure events to estimate the marginal mortality effect of a unit shock to network distance, exploiting within-municipality variation to bypass the cross-sectional confounders identified here.

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A Robustness

This appendix reports the robustness battery for the embedding and the divergence metric. All numbers are computed on the same data as the main text and re-run from scratch whenever the random seed or hyperparameter changes; the numerical results below come from the `10_robustness.py` script and are reproducible by replaying it on the public replication archive.

Figure 8 summarizes the four panels of robustness that follow.

Seed stability. We re-train the bipartite embedding under three additional seeds ($\{1, 13, 100\}$ in addition to the baseline 42) holding hyperparameters fixed, and compute the Pearson correlation between embedding distance under the baseline and each alternative seed on 5,000 municipality pairs randomly selected with a fixed seed (so that the same pairs are queried across all comparisons). The correlations are 0.71 (seed 1), 0.71 (seed 13), and 0.70 (seed 100). The top-10 embedding-cosine neighbors of São Paulo overlap with the baseline list in 3–5 out of 10 positions; for Manaus, 3–4 out of 10. Although individual neighbor lists are not identical across seeds, the substantive geographic signal is preserved: the metropolitan ABC and the Manaus ribeirinho cluster always dominate the

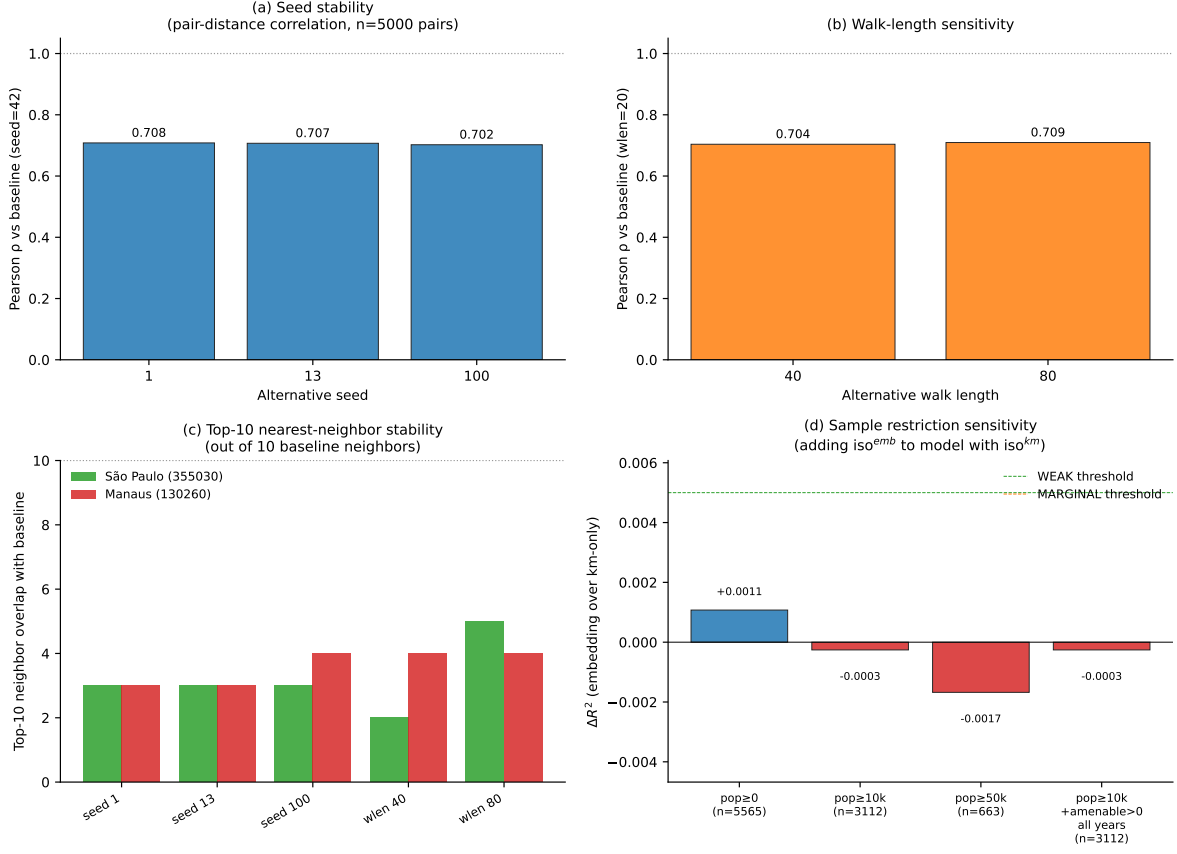


Figure 8: Robustness battery for the embedding and the predictive null. (a) Pearson correlation of pair-distance under three alternative seeds vs. baseline. (b) Same for two alternative walk lengths. (c) Top-10 nearest-neighbor overlap with baseline for São Paulo and Manaus. (d) ΔR^2 from adding embedding isolation to the geographic-isolation model, by sample restriction.

respective top-10, only the relative ordering of municipalities within each cluster is unstable. The 0.7 pair-distance correlation indicates that the random seed captures genuine specification uncertainty in the embedding; the divergence map therefore should be read at the level of regional patterns, not at the level of any individual municipality’s exact rank.

Walk-length sensitivity. We re-train the embedding with walk length $\ell \in \{40, 80\}$ in addition to the baseline $\ell = 20$, holding seed and other hyperparameters fixed. Pair-distance correlations against baseline are 0.70 ($\ell = 40$) and 0.71 ($\ell = 80$). The $\ell = 80$ specification yields slightly higher top-10 overlap with baseline (5/10 for São Paulo, 4/10 for Manaus) than $\ell = 40$ (2/10, 4/10). The correlation pattern suggests that the dominant source of embedding variation is the random walk realization, not the walk length, once $\ell \geq 20$.

Alternative high-complexity definitions. The baseline defines a high-complexity hospital as any hospital with an active SUS habilitation in any of seven categories. We test two alternatives:

(i) *Narrow definition: oncology and cardiology only.* Restricting the high-complexity set to hospitals holding habilitations in ONCO or CARDIO ($n = 4,725$ hospitals), the Spearman rank-correlation of the resulting iso^{emb} measure against the baseline is 0.33. This is a substantial decrease, indicating that the choice of high-complexity definition matters for the isolation metric. The narrow definition shrinks the eligible hospital set by roughly 50% and many municipalities switch nearest hospitals as a consequence.

(ii) *Long-lived habilitations: ≥ 24 months active.* Restricting to hospitals whose habilitation was active in at least 24 months during the panel window screens out short-lived administrative classifications. The resulting iso^{emb} has Spearman 0.92 with baseline. The choice of activity-duration threshold is relatively innocuous.

The implication is that the network-revealed isolation measure is robust to the activity-duration cut but sensitive to the categorical breadth of the high-complexity definition. A reader interpreting the divergence map should bear in mind that the metric reflects access to the specific subset of services covered by the included habilitations; applications targeting specific clinical conditions should re-derive the metric over the relevant restricted hospital set.

Sample restriction sensitivity. The mortality regression in Section 5.7 uses the sample of municipalities with mean population $\geq 10,000$, yielding $n = 3,112$ observations. We re-run the regression on three additional samples:

- $n = 5,565$ (the unrestricted sample): ΔR^2 from adding embedding-isolation to the geographic-isolation model is +0.001.
- $n = 663$ (mean population $\geq 50,000$): $\Delta R^2 = -0.002$.
- $n = 3,112$ (mean population $\geq 10,000$ and non-zero amenable death count in every year of the panel): $\Delta R^2 = -0.0003$ (statistically indistinguishable from the baseline restriction).

The qualitative null is robust across these samples. Notably, the sign of the ΔR^2 flips slightly positive when small municipalities are included; we interpret this as an artifact of the SIM under-reporting gradient discussed in Section 6: the embedding captures sub-population structure that, combined with the mechanical relationship between observed amenable mortality and SIM-coverage gradients, produces a small predictive gain in the smallest municipalities. This pattern strengthens rather than weakens the methodological warning: the cross-sectional regression is informative about observed amenable mortality, not about the underlying mortality the access measure should predict.

Table 5: State-level miscalibration of the SUS hospital network. States ordered by mean $|\tilde{\Delta}|$ (highest = least calibrated). Sign of mean $\tilde{\Delta}$ indicates whether the typical municipality in the state is more isolated than km suggests (+) or less (-). Last column reports the share of the state’s municipalities with $|\tilde{\Delta}| > 1$.

UF	State	Reg	<i>n</i> muns	$ \tilde{\Delta} $	$\tilde{\Delta}$	km med	ext+	ext-	ext%
DF	Distrito Federal	CO	1	3.22	+3.22	0	1	0	100%
RR	Roraima	N	15	1.48	-1.26	134	0	10	67%
AC	Acre	N	22	1.21	-1.00	81	1	13	64%
AP	Amapá	N	16	1.14	-0.87	70	1	9	62%
SE	Sergipe	NE	75	1.11	-0.84	19	2	43	60%
AL	Alagoas	NE	102	1.01	-0.76	18	4	49	52%
TO	Tocantins	N	139	0.91	-0.51	36	6	52	42%
GO	Goiás	CO	246	0.87	+0.48	19	59	18	31%
PB	Paraíba	NE	223	0.86	-0.45	16	14	78	41%
MT	Mato Grosso	CO	141	0.84	-0.15	53	20	27	33%
RN	Rio Grande do Norte	NE	167	0.82	-0.52	17	2	60	37%
MA	Maranhão	NE	217	0.75	+0.45	20	43	8	24%
ES	Espírito Santo	SE	78	0.75	-0.07	10	9	16	32%
PI	Piauí	NE	224	0.73	-0.16	39	20	36	25%
SP	São Paulo	SE	645	0.73	+0.39	13	126	37	25%
PE	Pernambuco	NE	185	0.73	-0.01	11	15	24	21%
BA	Bahia	NE	417	0.70	+0.25	23	67	28	23%
PR	Paraná	S	399	0.68	+0.08	19	47	40	22%
MG	Minas Gerais	SE	853	0.67	+0.03	17	80	94	20%
MS	Mato Grosso do Sul	CO	78	0.66	+0.08	0	7	9	21%
SC	Santa Catarina	S	293	0.65	-0.11	11	22	36	20%
RJ	Rio de Janeiro	SE	92	0.65	+0.43	0	14	4	20%
AM	Amazonas	N	62	0.64	+0.19	47	9	6	24%
CE	Ceará	NE	184	0.64	-0.25	0	4	44	26%
PA	Pará	N	143	0.63	+0.07	0	12	21	23%
RS	Rio Grande do Sul	S	496	0.58	-0.22	13	11	72	17%
RO	Rondônia	N	52	0.52	+0.13	32	3	3	12%

Notes: $|\tilde{\Delta}|$ is the mean of the unsigned divergence *z*-score; higher means the state’s municipalities deviate more from geographic prediction in either direction. $\tilde{\Delta}$ is the signed mean; ext+ (resp. ext-) counts municipalities with $\tilde{\Delta} > 1$ (resp. < -1). km med is the median geographic distance to the nearest high-complexity hospital, in kilometers.

Table 6: Out-of-sample R^2 for amenable mortality regressions, ridge with 5-fold CV.

Specification	R^2	SD	MAE	ΔR^2 vs km
M_0 : pop + GDP + UF FE	0.392	0.056	26.5	–
M_{km} : $M_0 + z(\log \text{iso}^{km})$	0.405	0.055	26.0	–
M_{emb} : $M_0 + z(\text{iso}^{emb})$	0.397	0.054	26.3	-0.009
M_{both} : $M_0 + z(\log \text{iso}^{km}) + z(\text{iso}^{emb})$	0.405	0.055	26.0	-0.000
$M_{ \tilde{\Delta} }$: $M_0 + \tilde{\Delta} $	0.392	0.055	26.5	-0.013
$M_{km+ \tilde{\Delta} }$: $M_0 + z(\log \text{iso}^{km}) + \tilde{\Delta} $	0.406	0.056	26.0	+0.001

Notes: Sample is $n = 3,112$ municipalities with mean population $\geq 10,000$, 2010–2023. Outcome is the period-pooled amenable mortality rate per 100,000, winsorized at [1%, 99%]. Ridge penalty $\alpha = 1$, random seed 42.